

Enterprise Land-to-Development Intelligence Platform: Architectural Blueprint, Regulatory Engine, and Computational Feasibility Framework

Deliverable 1: Planwise Source Truth Report

An exhaustive reverse-engineering audit of the eighty-seven pages of the *Planwise Site Development Masterclass* establishes the empirical boundary between proven source facts, operational inferences, statutory municipal baselines, and unresolved technical implementations. In conventional practice, real estate land acquisition feasibility requires four to six weeks of fragmented manual coordination among land surveyors, municipal planning consultants, architects, and financial analysts. The Planwise platform compresses this timeline down to minutes by binding client-side WebGL canvas drawing directly to property boundaries, spatial exclusions, and planning parameter forms.

To build an enterprise-grade platform that avoids unverified assumptions, every architectural claim, functional capability, and technical constraint is classified under five explicit epistemic categories:

- **A. DOCUMENTED:** Explicitly stated, demonstrated, or provided with code and specifications within the reference masterclass document.
- **B. REASONABLY INFERRED:** Logically necessary derived implementation details following standard software and geospatial engineering patterns.
- **C. EXTERNALLY VERIFIED:** Confirmed by authoritative statutory documents, municipal building bye-laws, or standard geospatial library behaviors.
- **D. PROPOSED / RECOMMENDED:** Modern enterprise design patterns, optimized algorithms, and scalable architectures designed to transcend prototype limitations.
- **E. UNKNOWN:** System details omitted from the primary text with no determinable evidence.

Planwise Source Truth Matrix

Domain Area	Document Evidence	Epistemic Status	Confidence	Missing Information
Product Purpose	Compresses land acquisition feasibility from 4–6 weeks to minutes by digitizing boundaries and evaluating spatial constraints.	A. DOCUMENTED	High	Integration pathways to municipal automated building approval portals (e.g., AutoDCR).
Target Users	Real estate	A. DOCUMENTED	High	Multi-tenant

Domain Area	Document Evidence	Epistemic Status	Confidence	Missing Information
	developers, master planners, architects, institutional land aggregators, and transaction advisory firms.			enterprise organization hierarchies, team permission boundaries, and client sharing models.
Business Workflow	Sequential lifecycle: Authentication → Dashboard → Project Init → Boundary Delineation → Feature Placement → Constraint Entry → Summary Review.	A. DOCUMENTED	High	Specific state machine handling for branched project revisions or multi-parcel mergers.
User Journeys	End-to-end journey from /login through /home, /project-selection, and /map to /site-constraints.	A. DOCUMENTED	High	Rollback handling when navigation occurs prior to spatial persistence completion.
Project Lifecycle	Projects encapsulate title, description, land ownership type, primary GeoJSON boundary, deductions, and tree markers.	A. DOCUMENTED	High	Project archiving states, historical version branches, and soft-delete retention rules.
UI Architecture	Single Page Application (SPA) with floating tool palettes, heads-up metrics display (HUD), and modal form dialogs.	A. DOCUMENTED	High	Responsive multi-breakpoint layouts for mobile/tablet site inspectors.
React Architecture	React 18 functional	A. DOCUMENTED	High	Global state management

Domain Area	Document Evidence	Epistemic Status	Confidence	Missing Information
	components, hooks (useState, useRef, useCallback, useEffect), and React Router v6.			configuration (Zustand, Redux) or server-state caching.
Mapbox Engine	Mapbox GL JS coupled with @mapbox/mapbox-gl-draw, custom SVG controls, and Mapbox Geocoder restricted to India.	A. DOCUMENTED	High	Custom WebGL shader implementations for real-time shadow casting and terrain drape.
GeoJSON Architecture	RFC 7946 WGS84 GeoJSON coordinate arrays for boundaries (Polygon), deductions (Polygon), and trees (Point).	A. DOCUMENTED	High	Mathematical projection transformations from EPSG:4326 degrees to local planar metric coordinates.
API Contracts	REST endpoints (/api/projects), JSON payloads with stringified geometry fields, and Bearer JWT authorization.	A. DOCUMENTED	High	OpenAPI schemas, rate limiting, request validation middleware, and idempotency keys.
Authentication Flow	Email/password submission returning signed JWT; token persisted in browser localStorage and injected into Axios headers.	A. DOCUMENTED	High	Refresh token rotation, multi-factor authentication, and session invalidation mechanisms.
Database Tier	Relational schemas referenced; Supabase and PostgreSQL mentioned in bibliography.	B. REASONABLY INFERRED	Low	Exact DBMS engine, spatial indexes (GIST), table schemas, foreign keys, and migration scripts.
Spatial	Geodesic area	A. DOCUMENTED	Medium	Inward setback

Domain Area	Document Evidence	Epistemic Status	Confidence	Missing Information
Operations	calculation, ring closure verification, and basic point-in-polygon containment checks.			offsetting, straight skeleton generation, and boolean polygon clipping.
Regulatory Inputs	Manual form entry for abutting road width, front/rear/side setbacks, height limits, and land ownership enums.	A. DOCUMENTED	High	Dynamic rule engine capable of parsing contradictory statutory clauses.
Business Constraints	Captures commercial-to-residential ratios, target carpet areas, and parking typology (Basement vs. Podium).	A. DOCUMENTED	High	Unit-mix distribution ratios, floor-to-floor heights, and retail depth constraints.
Feasibility Workflow	Computes gross plot area and subtracts non-developable zones and tree buffers to show net developable footprint.	A. DOCUMENTED	High	Algorithmic resolution of buildable envelopes on non-convex or acute-angled plots.
Layout Workflow	Mentions downstream generative layout tools for structural column grids, corridors, and 1BHK/2BHK/3BHK units.	E. UNKNOWN	Low	Algorithmic implementation of generative layout synthesis (e.g., MILP vs. Wave Function Collapse).
Massing Workflow	Mentions extrusion of 2D floor plates into 3D structural volumes across podiums, transfer slabs, and tower envelopes.	B. REASONABLY INFERRED	Low	Procedural 3D mesh generation, step-back offsets, and aerodynamic corner rounding.

Domain Area	Document Evidence	Epistemic Status	Confidence	Missing Information
Testing Strategy	Manual debugging using Chrome DevTools, console error monitoring, and network payload inspection.	A. DOCUMENTED	High	Automated unit tests, geospatial fuzzing suites, visual regression, and CI/CD pipelines.
Deployment Setup	Local development environment running on localhost:3000 (React SPA) and localhost:5001 (API service).	A. DOCUMENTED	High	Cloud infrastructure, Kubernetes manifests, container definitions, and CDN caching.
Security Model	Route guarding via <ProtectedRoute> reading localStorage token; vulnerable to cross-site scripting (XSS).	A. DOCUMENTED	Medium	Enterprise security: HttpOnly cookies, CSRF protection, CSP directives, and encryption.
System Boundaries	Confined strictly to early-stage pre-feasibility analysis without municipal approval workflows.	A. DOCUMENTED	High	Formal integration points with local planning authorities and licensed architects.
Inferred Architecture	Node.js/Express REST backend on port 5001 parsing stringified GeoJSON payloads into document/relational stores.	B. REASONABLY INFERRED	Medium	Complete backend framework specifications, process supervisors, and connection pool sizes.
Missing Geometry	Straight skeleton inward offsets, usable rectangle extraction, and medial axis transforms.	D. PROPOSED / RECOMMENDED	High	Exact implementation algorithms required for production geometric compliance.
Missing Financials	Comprehensive real estate financial	D. PROPOSED / RECOMMENDED	High	Full institutional investment formulas (IRR,

Domain Area	Document Evidence	Epistemic Status	Confidence	Missing Information
	pro-formas, cash-flow phasing, debt schedules, and sensitivity models.			NPV, Residual Land Value, Equity Multiple).
Missing Regulations	Versioned, machine-readable rule engine encoding state and municipal development control regulations.	D. PROPOSED / RECOMMENDED	High	Structured rule schemas for DCPR 2034, UDCPR, RMP 2015/2025, MPD 2021, and JBBL 2016.

Deliverable 2: Complete Product Architecture

The proposed system addresses the limitations of the prototype by establishing an enterprise-grade platform organized into twelve functional tiers. This hierarchy decouples user interactions from deterministic geometric computations, statutory rule resolutions, and generative design optimizations.

Architectural Hierarchy and Tier Decomposition

- **Level 1: Project Management:** Orchestrates multi-tenant workspaces, enterprise organizations, parcel revision histories, and granular Role-Based Access Control (RBAC). It enables collaborative editing across acquisition teams, master planners, and financial analysts using Conflict-Free Replicated Data Types (CRDTs) to prevent conflicting updates during concurrent site revisions.
- **Level 2: Site Intelligence:** Coordinates external geospatial data pipelines. It ingests cadastral survey sheets, land title records, high-resolution satellite imagery, digital elevation models (DEMs), and municipal Development Plan (DP) maps. The engine derives digital contours, surface hydrology, slope gradients, and macro-infrastructure connectivity.
- **Level 3: Regulatory Intelligence:** A deterministic, machine-readable rule compilation engine. It ingests versioned municipal development control regulations (e.g., Mumbai DCPR 2034, Maharashtra UDCPR 2020/2025, Bengaluru RMP 2015/2025-2026, Delhi MPD 2021, and Jharkhand JBBL 2016). It programmatically evaluates road-width thresholds, base and gross Floor Space Index (FSI), height restrictions, and statutory surrender reservations.
- **Level 4: Geometry / Buildable Envelope Engine:** Transforms geographic WGS84 coordinates into local metric planar projections (UTM). It executes boolean polygon clipping for road-widening reservations, environmental buffers, and utility easements, and applies inward Minkowski setbacks and straight skeletons to construct the exact 3D buildable volume.
- **Level 5: Generative Site Planning:** Synthesizes macro site layouts across single-plot or multi-parcel assemblies. It optimizes building footprint positions, podium bases, open space reservations, pedestrian plazas, and perimeter vehicular circulation loops meeting

mandatory fire tender movement norms.

- **Level 6: Building Layout Engine:** Procedurally generates internal architectural floor plates within the buildable footprint. It places structural elevator and stair cores, egress corridors, MEP service shafts, and residential unit partitions (1BHK through 4BHK) adhering strictly to National Building Code (NBC 2016) habitable room dimensions and light/ventilation requirements.
- **Level 7: 3D Massing Engine:** Transforms 2D floor plates into parametric 3D structural meshes. It models step-back terraces, podium-to-tower transfer structures, mechanical service floors, and rooftop amenities, rendering the resulting masses within an interactive WebGL viewport.
- **Level 8: Parking / Mobility Engine:** Automates the layout of subterranean basements, ground stilts, and multi-level podium parking decks. It packs standard vehicle bays, designs two-way driveways, calculates ramp slopes adhering to statutory gradient limits, validates fire tender turning sweeps, and allocates mandatory electric vehicle (EV) charging infrastructure.
- **Level 9: Financial Feasibility Engine:** A real estate development pro-forma engine that models project financial viability. It evaluates Gross Development Value (GDV), Total Development Cost (TDC), hard and soft construction costs, statutory authority approval premiums, Transferable Development Rights (TDR) purchases, financing interest, and sales milestones, deriving project NPV, IRR, and Residual Land Value (RLV).
- **Level 10: Scenario / Optimization Engine:** Implements a Multi-Objective Evolutionary Algorithm (NSGA-II) to balance conflicting development priorities. It evaluates trade-offs between maximizing saleable carpet area, minimizing construction cost, optimizing daylight exposure, and reducing structural complexity, generating a Pareto frontier of viable development configurations.
- **Level 11: AI Development Copilot:** A hybrid intelligence layer that combines Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) over municipal planning documents. The copilot provides natural language querying and regulatory explanations. It operates behind deterministic validation gates, ensuring the AI can query and explain regulations but cannot invent or alter geometric or financial values.
- **Level 12: Reporting / Collaboration:** Compiles comprehensive feasibility prospectuses, board investment memorandums, vector CAD/BIM exports (Industry Foundation Classes [IFC] and DXF), and executive summary dossiers with full statutory calculation audit trails.

Deliverable 3: Mathematical / Formula Handbook

To ensure legal defensibility and precision across engineering calculations, the platform operates on strict mathematical foundations. All internal spatial computations use SI units (meters, square meters, radians), converting to regional real estate units (square feet, acres, gunthas) only at the presentation boundary.

Distance, Bearing, and Geodesic Formulas

- **Vincenty's Inverse Formula (Geodesic Distance):** Calculates the shortest distance over the WGS84 reference ellipsoid ($a = 6378137.0 \text{ m}$, $b = 6356752.314245 \text{ m}$, $f = 1/298.257223563$):
$$\tan U_1 = (1 - f) \tan \phi_1, \quad \tan U_2 = (1 - f) \tan \phi_2$$
$$\lambda = L \quad (\text{iterated until } |\Delta \lambda| < 10^{-12})$$
$$\sin \sigma = \sqrt{(\cos$$

$$U_2 \sin \lambda)^2 + (\cos U_1 \sin U_2 - \sin U_1 \cos U_2 \cos \lambda)^2 \cos \sigma = \sin U_1 \sin U_2 + \cos U_1 \cos U_2 \cos \lambda, \quad \sigma = \arctan2(\sin \sigma, \cos \sigma)$$

$$\sin \alpha = \frac{\cos U_1 \cos U_2 \sin \lambda}{\sin \sigma}, \quad \cos^2 \alpha = 1 - \sin^2 \alpha \cos(2\sigma_m) = \cos \sigma - \frac{2 \sin U_1 \sin U_2 \cos^2 \alpha}{s} = b \cdot B \cdot (\sigma - \Delta \sigma)$$
Precision: 0.001 mm; *Assumptions:* Non-antipodal points on WGS84.

- Forward Geodesic Initial Bearing:** $\theta = \arctan2(\sin(\Delta \lambda) \cos \phi_2, \cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos(\Delta \lambda)) \pmod{2\pi}$ *Units:* Radians or decimal degrees clockwise from True North.

Planar and Spherical Area Formulas

- Planar Shoelace Formula (Green's Theorem Discrete Form):** $A_{\text{planar}} = \frac{1}{2} \left| \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right|$ *Units:* Square meters (m^2); *Assumptions:* Simple, non-self-intersecting planar polygon in projected metric coordinates ($x_n = x_0, y_n = y_0$).
- Spherical Excess Area (Girard's Theorem for Geodesic Polygons):** $E = \sum_{j=1}^k \alpha_j - (k - 2)\pi$, $A_{\text{spherical}} = R^2 \cdot E$ Where α_j are the interior spherical vertex angles and $R = 6,371,008.8 \text{ m}$ (mean authalic radius).

Financial Mathematics and Discounted Cash Flows

- Net Present Value (NPV):** $\text{NPV} = \sum_{t=0}^T \frac{CF_t}{(1 + r)^t}$ Where CF_t represents the net cash flow in month/year t , and r is the periodic discount rate.
- Internal Rate of Return (IRR):** The discount rate that drives net present value to zero: $\sum_{t=0}^T \frac{CF_t}{(1 + \text{IRR})^t} = 0$
- Exact Day-Count Internal Rate of Return (XIRR):** Solves for the annualized internal rate of return across irregularly spaced cash flows: $\sum_{i=1}^N \frac{CF_i}{(1 + \text{XIRR})^{\frac{d_i - d_1}{365}}} = 0$ Where d_i is the exact transaction date of the i -th cash flow and d_1 is the baseline investment date.
- Residual Land Value (RLV):** $\text{RLV} = \frac{\text{GDV}}{C_{\text{statutory}}} + C_{\text{hard}} + C_{\text{soft}} + C_{\text{finance}} + C_{\text{contingency}} \{1 + M_{\text{target_profit}}\} - C_{\text{stamp_duty}}$ Where $M_{\text{target_profit}}$ is the developer's required margin on cost (TDC).

Deliverable 4: Computational Geometry Specification

The platform implements thirty-eight fundamental geometric operations. The engine prioritizes exact topological calculations over graphical approximations, using planar metric projections (UTM) for statutory calculations and re-projecting to WGS84 only for browser cartography.

Master Computational Geometry Operations Matrix

#	Operation	Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation	PostGIS / Python / Frontend Implementation
1	Planar Polygon Area	$A = \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i)$	Shoelace / Green's Theorem. $\mathcal{O}(n)$.	Collapsed zero-width corridors; self-intersections invalidate result.	ST_Area(geom) / shapely.area / Turf.js area
2	Spherical Geodesic Area	$E = \sum \alpha_j - (k-2)\pi$; $A = R^2 E$	Spherical excess integration. $\mathcal{O}(n)$.	Antimeridian crossings; pole inclusions.	ST_Area(geom, true) / pyproj.Geod / Custom Karney integration
3	Projected Planar Area	Area computed after conformal map projection (UTM).	Transverse Mercator scale correction. $\mathcal{O}(n)$.	Grid scale distortion away from central meridian.	ST_Area(ST_Transform(geom, srid)) / pyproj.Transformer
4	Perimeter	$P = \sum \sqrt{\Delta x_i^2 + \Delta y_i^2}$	Euclidean summation. $\mathcal{O}(n)$.	Duplicate consecutive vertices must be pruned first.	ST_Perimeter(geom) / shapely.length / Turf.js length
5	Minimum Bounding Box	$\min_{\theta} (W_{\theta} \times H_{\theta})$ containing polygon.	Rotating Calipers over convex hull. $\mathcal{O}(n \log n)$.	Collinear edges; degenerate lines.	ST_OrientedEnvelope(geom) / shapely.minimum_rotated_rectangle
6	Geometric Centroid	$C = \frac{1}{6A} \sum (p_i + p_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$	Second moment of area integral. $\mathcal{O}(n)$.	Centroid may fall outside concave or horseshoe polygons.	ST_Centroid(geom) / shapely.centroid / Turf.js centroid
7	Polygon Orientation	$\text{sgn} \left(\sum (x_{i+1} - x_i)(y_{i+1} + y_i) \right)$	Contour trapezoidal summation. $\mathcal{O}(n)$.	CCW for exterior rings; CW for interior holes.	ST_ForcePolygonCCW(geom) / shapely.orient
8	Ring Closure	$v_0 = v_n$ iff $(x_0 = x_n \wedge y_0 = y_n)$	Structural assertion. $\mathcal{O}(1)$.	Open LineString mistakenly treated as a Polygon.	ST_AddPoint(line, ST_StartPoint(line)) / Python assertion
9	Self-Intersection Detection	$\exists s_i, s_j$ s.t. $s_i \cap s_j \neq \emptyset$	Bentley-Ottman sweep-line. $\mathcal{O}(n \log n)$.	Bowtie polygons create invalid	ST_IsSimple(geom) / ST_IsValid(geom)

#	Operation	Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation	PostGIS / Python / Frontend Implementation
		$\emptyset \text{ and } \text{Vert}\{i-j\} \text{Vert}\{i-j\} > 1$	$+k)\log n)$.	topological interiors.	$m) / \text{shapely.is_valid}$
10	Polygon Validity	Enforces OGC simple feature specifications.	GEOS topological rule suite. $\mathcal{O}(n \log n)$.	Self-touching interior rings; nested holes.	ST_IsValidReason(geom) / shapely.is_valid
11	Polygon Union	$P_A \cup P_B = \{p \mid p \in P_A \vee p \in P_B\}$	Martinez-Rueda-Feito boolean union. $\mathcal{O}(n \log n)$.	Disjoint polygons return a MultiPolygon.	ST_Union(geom_a, geom_b) / shapely.union
12	Polygon Intersection	$P_A \cap P_B = \{p \mid p \in P_A \wedge p \in P_B\}$	Weiler-Atherton / Martinez-Rueda. $\mathcal{O}(n \log n)$.	Coplanar edges generate degenerate linear slivers.	ST_Intersection(geom_a, geom_b) / shapely.intersection
13	Polygon Difference	$P_A \setminus P_B = \{p \mid p \in P_A \wedge p \notin P_B\}$	Boolean clipping graph traversal. $\mathcal{O}(n \log n)$.	Subtractions that split a polygon into disjoint islands.	ST_Difference(geom_a, geom_b) / shapely.difference
14	Polygon Clipping	$P \cap \text{Bounding Window}$	Sutherland-Hodgman / Liang-Barsky. $\mathcal{O}(n)$.	Concave clipping windows create multi-part outputs.	ST_ClipByBox2D(geom, box) / shapely.clip_by_rect
15	Offset Curves	Swept envelope along boundary normal vectors.	Minkowski sum / circular arc expansion. $\mathcal{O}(n)$.	Miter limits balloon sharp acute corners.	ST_OffsetCurve(line, d) / shapely.offset_curve
16	Buffer Generation	$B(P, d) = \{x \mid \text{dist}(x, P) \leq d\}$	Minkowski sum with disk of radius d. $\mathcal{O}(n \log n)$.	Large buffers merge nearby disjoint features.	ST_Buffer(geom, d) / shapely.buffer / Turf.js buffer
17	Negative Inward Buffer	$B(P, -d) = P \setminus B(\partial P, d)$	Interior wavefront propagation. $\mathcal{O}(n \log n)$.	Over-buffering collapses the polygon entirely.	ST_Buffer(geom, -d) / shapely.buffer(-d)

#	Operation	Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation	PostGIS / Python / Frontend Implementation
18	Setback Generation	$P \setminus \bigcup \text{HalfPlane}(e_i, d_i)$	Straight skeleton edge offsetting. $\mathcal{O}(n \log n)$.	Non-uniform edge setbacks create interior gaps.	Straight skeleton algorithm / CGAL Polygon Offset
19	Road Widening Deduction	$P_{\text{site}} \setminus \bigcup \text{Buffer}(\text{Centerline}, W_{\text{new}}/2)$	Boolean difference along frontage.	Frontage curves intersect irregular parcel corners.	ST_Difference(site, road_widening_poly)
20	Tree Buffer Generation	$B(p_{\text{tree}}, r_{\text{root}})$	Circular Minkowski disk generation.	Tree buffers crossing parcel boundaries.	ST_Bu[span_36](start_span)[span_36](end_span)ffer(point, radius) / shapely.buffer
21	Easement Deduction	$P_{\text{site}} \setminus \bigcup \text{Easement Polygons}$	Boolean subtraction of rights-of-way.	Easements bisecting parcels into disjoint zones.	ST_Difference(site, easements)
22	Non-Developable Subtraction	$P_{\text{site}} \setminus \bigcup \text{Exclusion Zones}$	Boolean subtraction of streams, slopes, lines.	Multi-ring outputs; interior island generation.	ST_Difference(site, non_dev_poly)
23	Buildable Envelope	Intersection of setback and deduction profiles.	Comprehensive spatial difference pipeline.	Resulting buildable area is zero or unusable.	Geometric pipeline: $\mathcal{P} \setminus \bigcup \text{Setbacks} \setminus \bigcup \text{Deductions}$
24	Fragmentation Analysis	Filter disconnected components by size: $\{P_i \mid A(P_i) \geq A_{\min}\}$.	Connected-component analysis. $\mathcal{O}(k)$.	Tiny slivers retained as valid buildable areas.	ST_Dump(envelope) filtered by area threshold

#	Operation	Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation	PostGIS / Python / Frontend Implementation
25	Minimum Frontage Analysis	Maximum continuous segment length along road edge.	Linear projection along frontage edges. $\mathcal{O}(n)$.	Curved frontages; broken, segmented frontages.	ST_Length(ST_Intersection(site, road))
26	Minimum Parcel Width	Minimum diameter across the buildable envelope.	Rotating calipers / medial axis inscribed circles.	Narrow bottlenecks dividing wider building areas.	Chaikin thinning or inscribed circle scan
27	Minimum Parcel Depth	Normal distance from front boundary to rear edge.	Perpendicular ray casting across centroid. $\mathcal{O}(n)$.	Triangular plots where depth tapers to zero.	Ray-edge intersection analysis
28	Access Corridor Width	Clear width along vehicular entry points.	Swept-sphere or parallel line buffering.	Gates, guardhouses, or columns obstructing access.	Inscribed corridor cross-section checks
29	Fire Access Corridor	Continuous minimum 6.0 m driveways around towers.	Dilation and erosion morphological filtering.	Driveways blocked by landscape walls or steps.	Buffer-erosion path clearance check
30	Turning Radius Validation	Swept path envelope for a 9.0 m [start_span][end_span] [start_span][end_span] [start_span][end_span] [start_span][end_span] radius fire tender.	Reeds-Shepp or Dubins vehicle path modeling.	Tight 90° corners pinching vehicular sweeps.	Swept-polygon intersection along turn paths
31	Usable Rectangle Extraction	$\max(w \cdot h) \text{ s.t. } R(w, h, \theta) \subseteq$	Largest inscribed rectangle branch-and-bo	Non-convex shapes yield multiple local optima.	Discrete raster grid scan or CGAL LIR solver

#	Operation	Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation	PostGIS / Python / Frontend Implementation
		$P_{\text{buildable}}$			
32	Irregular Decomposition	Partition irregular polygon into convex sub-regions.	Hertel-Mehlhorn convex partitioning. $\mathcal{O}(n \log n)$.	Over-partitioning generates excessive narrow shapes.	CGAL optimal_convex_partition_2
33	Convex Hull	Smallest convex set enclosing parcel: $\text{Conv}(P)$	Monotone Chain (Andrew's algorithm). $\mathcal{O}(n \log n)$.	Collinear vertices along boundary edges.	ST_ConvexHull(geom) / shapely.convex_hull
34	Concave Hull (Alpha Shape)	Minimal boundary enclosing points with detail level α .	Delaunay triangulation with edge radius filtering.	Overly small α splits the shape into islands.	ST_ConcaveHull(geom, target) / libpysal
35	Medial Axis / Skeleton	Topological skeleton equidistant to boundary edges.	Voronoi diagram of densified boundary points.	Boundary noise creates spurious skeleton branches.	Boost.Polygon Voronoi / scikit-image
36	Visibility (Isochrone) Analysis	Visible 2D polygon from an observer point p.	Angular plane sweep over vertex obstacles. $\mathcal{O}(n \log n)$.	Ray-surface grazing singularities at corners.	Ray-casting sweep over internal obstacles
37	Topological Adjacency	Checks whether boundaries share edges: $\dim(A \cap B) = 1$.	DE-9IM intersection matrix evaluation. $\mathcal{O}(n \log n)$.	Shapes touching only at an isolated point vertex.	ST_Touches(geom_a, geom_b) / shapely.touches
38	Connected Components	Grouping contiguous parcel elements into clusters.	Disjoint-set data structure (Union-Find). $\mathcal{O}(n)$.	Disjoint parcels separated by tiny survey gaps.	Graph connectivity over PostGIS adjacency queries

Deliverable 5: Regulation Engine Specification

The platform treats municipal planning codes as declarative, version-controlled software artifacts. Regulations are never hardcoded into UI views or calculation controllers. Instead, they are parsed by an independent Rule Engine that evaluates parcels against applicable statutes, maintains historical versions, and produces an auditable legal citation for every calculation.

Multi-Jurisdictional Statutory Reference Matrix

State / Region	Planning Authority	Statutory Code & Table	Base FSI	Maximum FSI	Road Width Thresholds	Statutory Setbacks & Rules
Maharashtra (Mumbai)	MCGM	DCPR 2034, Reg 30, Table 12	Island: 1.33 Suburbs: 1.00	Up to 2.70 (Resi) Up to 5.00 (Comm)	<9\text{ m}, 9\text{--}12\text{ m}, 12\text{--}18\text{ m}, 18\text{--}27\text{ m}	Premium FSI (50% ASR) + TDR based on road width. High-rise setbacks (scale with height).
Maharashtra (Rest of State)	Pune (PMC), Thane, Nagpur	UDCPR 2020 (Updated 2025), Table 6-A/6-G	1.10	Up to 3.00 (Standard) Up to 4.00 (TOD)	<9\text{ m}, 9\text{--}12\text{ m}, 12\text{--}15\text{ m}, 15\text{--}18\text{ m}, 18\text{--}24\text{ m}	Maximum building potential = $\text{Base} + \text{Pr}(\text{start_span})(\text{end_span})[\text{start_span}][\text{end_span}](\text{start_span})(\text{end_span})[\text{start_span}][\text{end_span}]\text{emium} + \text{TDR}$. Table 6-A for congested gaathan cores.
Karnataka (Bengaluru)	BBMP / GBA, BDA	RMP 2015 / 2025–2026 Zonal Amendments	1.75 to 3.25	+40% Premium FAR +20% TDR	<9\text{ m}, 9\text{--}12\text{ m}, 12\text{--}15\text{ m}, 15\text{--}18\text{ m}	Premium FAR charged at 50% guidance value. Setbacks follow Table 8 (\le

State / Region	Planning Authority	Statutory Code & Table	Base FSI	Maximum FSI	Road Width Thresholds	Statutory Setbacks & Rules
						12\text{ m} & Table 9.
Delhi NCR	DDA	MPD 2021 / Draft 2041, Table 4.2	2.00 (Group Housing)	Up to 3.00 (Purchasable FAR)	Frontage $\geq 18\text{ m}$; min plot $3,000\text{ m}^2$	Ground coverage max 33.3%. Building height capped by $H \leq 1.5 \times (\text{Road width} + \text{Setback})$.
Telangana (Hyderabad)	GHMC, HMDA	Telangana Building Rules, G.O. Ms. 168	Uncapped (Non-FSI city)	Governed by setbacks, coverage, and height	Road: $<9\text{ m}$ (10\text{ m} cap), 9 m (15\text{ m} cap), $>30\text{ m}$ (Uncapped)	Dynamic setbacks: all-round open space $= \frac{\text{Height}}{3}$ (minimum 3.0\text{ m} to 16.0\text{ m}).
Tamil Nadu (Chennai)	CMDA	TNCDBR 2019, Rule 35 & 39	1.50 to 2.00	Up to 2.75 (Non-MSB) Up to 3.25+ (Premium FSI)	Bands: 9 m , 12 m , 15 m , 18 m	Premium FSI charged at 40%–50% guideline value. Front setback scales with road width.
West Bengal (Kolkata)	KMC	KMC Building Rules 2009 (Amended 2025), Table 3 & 5	1.75 to 2.50	Up to 3.50 (with mass transit / wide road loading)	Means of access: 5 m , 7 m , 9 m	Permissible height tied strictly to means of access: $H = 1.5 \times$

State / Region	Planning Authority	Statutory Code & Table	Base FSI	Maximum FSI	Road Width Thresholds	Statutory Setbacks & Rules
					9\text{--}12\text{ m}, 12\text{--}15\text{ m}, ext{[span_99](start_span)[span_99](end_span) m}, >15\text{ m}	(\text{Road[s pan_100](sta rt_span)[spa n_100](end_ span)[span_ 101](start_sp an)[span_10 1](end_span) } + \text{Setback }).
Jharkhand (Ranchi, Dhanbad)	RMC, MADA, RRDA	JBBL 2016 (Amended 2026), Table 15 & 16	1.50 to 2.00	Up to 2.50 (Resi) Up to 5.00 (Commercial)	Bands: <6.1\text{ m}, 6.1\text{--}7.5\text{ m}, 7.5\text{--}9.0\text{ m}, 9.0\text{--}12.0\text{ m}, >12.0\text{ m}	Road <6.1\text{ m} restricts non-high-rise construction; high-rise requires \ge 12\text{ m}.

Declarative Rule Engine Schema Architecture

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "jurisdiction": "IN-MH-MUM",
  "authority": "Municipal Corporation of Greater Mumbai",
  "regulationCode": "DCPR-2034",
  "version": "2034.4.2",
  "effectiveFrom": "2018-11-13",
  "effectiveTo": null,
  "citation": {
    "document": "Development Control and Promotion Regulations for  
Greater Mumbai, 2034",
    "clause": "Regulation 30, Table 12",
    "gazetteNotification": "No. TPB 4317/629/CR-118/2017/UD-11"
  },
  "applicability": {
    "zones": ["RESIDENTIAL_R2", "COMMERCIAL_C1"],
    "minPlotAreaSqm": 0,

```

```

    "maxPlotAreaSqm": null
  },
  "roadWidthRules": [
    {
      "minRoadWidthM": 0.0,
      "maxRoadWidthM": 9.0,
      "baseFsi": 1.00,
      "premiumFsiMax": 0.00,
      "tdrFsiMax": 0.00,
      "maxBuildingPotential": 1.00,
      "maxPermissibleHeightM": 15.0,
      "mandatoryFrontSetbackM": 3.0
    },
    {
      "minRoadWidthM": 9.0,
      "maxRoadWidthM": 12.0,
      "baseFsi": 1.00,
      "premiumFsiMax": 0.50,
      "tdrFsiMax": 0.50,
      "maxBuildingPotential": 2.00,
      "maxPermissibleHeightM": 32.0,
      "mandatoryFrontSetbackM": 4.5
    },
    {
      "minRoadWidthM": 12.0,
      "maxRoadWidthM": 18.0,
      "baseFsi": 1.00,
      "premiumFsiMax": 0.50,
      "tdrFsiMax": 0.70,
      "maxBuildingPotential": 2.20,
      "maxPermissibleHeightM": 50.0,
      "mandatoryFrontSetbackM": 6.0
    },
    {
      "minRoadWidthM": 18.0,
      "maxRoadWidthM": 27.0,
      "baseFsi": 1.00,
      "premiumFsiMax": 0.50,
      "tdrFsiMax": 0.90,
      "maxBuildingPotential": 2.40,
      "maxPermissibleHeightM": 70.0,
      "mandatoryFrontSetbackM": 6.0
    },
    {
      "minRoadWidthM": 27.0,
      "maxRoadWidthM": null,
      "baseFsi": 1.00,
      "premiumFsiMax": 0.50,

```

```

        "tdrFsiMax": 1.20,
        "maxBuildingPotential": 2.70,
        "maxPermissibleHeightM": 120.0,
        "mandatoryFrontSetbackM": 9.0
    }
],
"fireTenderAccess": {
    "heightThresholdM": 15.0,
    "minDrivewayClearanceM": 6.0,
    "minTurningRadiusM": 9.0,
    "maxDeadEndLengthM": 45.0
}
}

```

Deliverable 6: Generative Layout Engine Specification

The Generative Layout Engine synthesizes constructible floor plates within the 3D buildable envelope. The process follows a structured geometric dependency path:

- Stage 1: Ingest the buildable envelope derived from setbacks, easements, and road widenings.
- Stage 2: Generate the structural bay grid, aligning column lines to standard 6.0 m to 7.5 m spans.
- Stage 3: Place the vertical circulation cores (elevators, fire stairs, pressurized lobbies, MEP shafts).
- Stage 4: Route the central corridor backbone, enforcing a minimum width of 1.5 m to 2.0 m and capping dead ends at 6.0 m .
- Stage 5: Optimize residential unit demising walls, positioning them along structural column lines.
- Stage 6: Pack habitable rooms, kitchens, bathrooms, and utility risers to satisfy NBC 2016 requirements.

Statutory Residential Dimension Constraints (National Building Code 2016)

The engine enforces statutory dimensional minimums based on NBC 2016 (Part 3 & Part 4) to ensure generated layouts remain constructible and compliant with building codes:

- **Habitable Rooms (Bedrooms, Living Rooms):** Minimum clear floor area of 9.5 m^2 with a minimum horizontal width of 2.4 m `m][span_111](start_span)[span_111](end_span)[span_113](start_span)[span_113](end_span)`. In multi-room apartments, secondary bedrooms must measure at least 7.5 m^2 with a minimum width of 2.1 m . Minimum clear ceiling height is 2.75 m .
- **Kitchens:** Minimum clear area of 5.0 m^2 with a minimum width of 1.8 m when a separate dining area is provided; minimum 7.5 m^2 with a width of 2.1 m when combined with a dining area.
- **Bathrooms & Water Closets (W.C.):** Standalone water closets must measure at least 1.1 m^2 (minimum width 0.9 m); standalone bathrooms must measure at least

1.8 m^2 (minimum width 1.2 m); combined toilet-bathrooms must measure at least 2.8 m^2 (minimum clear width 1.2 m).

- **Balconies:** Minimum clear projection width of 1.0 m ; total projection must not exceed 1.2 m to 1.5 m into statutory setbacks, with linear runs capped by regional bye-law amendments.
- **Stairways and Exit Corridors:** Minimum clear corridor width of 1.5 m for residential apartment buildings (increasing to 2.0 m for commercial/mixed-use). High-rise buildings (height $>15\text{ m}$) require a minimum of two enclosed, fire-rated staircases with minimum flight widths of 1.5 m and positive-pressure air shafts. The maximum travel distance from any apartment entry to the nearest fire exit is 22.5 m (dead-end corridors capped at 6.0 m).

Deliverable 7: 3D Massing Engine Specification

The 3D engine converts 2D floor plates into interactive 3D building meshes, applying vertical setbacks, step-backs, and podium extrusions.

WebGL 3D Technology Comparison

Evaluation Dimension	Three.js (r160+)	Babylon.js (v6+)	Mapbox GL JS 3D (v3+)	CesiumJS (v1.115+)
Primary Domain	General WebGL 3D rendering, custom geometry generation, and shader pipelines.	In-browser game development, physically-based rendering (PBR), and complex material shaders.	Vector cartography, 2D basemaps, and satellite imagery draping.	Planetary-scale 3D GIS, global elevation terrain, and massive urban digital twins.
Geospatial Coordinate Support	None natively. Requires custom transformation matrices to map UTM/EPSG:3857 coordinates.	None natively. Requires custom Cartesian transformations.	Native EPSG:4326 and EPSG:3857 coordinate transformations on the GPU.	Native WGS84 ellipsoidal coordinates and OGC 3D Tiles streaming.
Procedural Extrusion Flexibility	High. Native ExtrudeGeometry with boolean CSG libraries.	High. Built-in procedural mesh builders and CSG plugins.	Moderate. Limited to single-polygon 3D extrusion layers (fill-extrusion).	Moderate. Supports 3D extrusions via CZML and batched 3D Tiles.
Runtime Memory Footprint	Low to Moderate ($15\text{--}40\text{ MB}$ base memory).	Moderate ($30\text{--}60\text{ MB}$ base memory).	Low to Moderate ($20\text{--}50\text{ MB}$ base memory).	High ($120\text{--}250\text{ MB}$ base memory).
Platform Selection Strategy	Core Selection for Generative Building Massing integrated via custom Mapbox	Retained for standalone interior unit rendering modules.	Core Selection for Basemap and Macro Site Spatial Context.	Recommended for regional master planning exceeding 100 km^2 .

Evaluation Dimension	Three.js (r160+)	Babylon.js (v6+)	Mapbox GL JS 3D (v3+)	CesiumJS (v1.115+)
	WebGL layers.			

WebGL Lifecycle and Custom Three.js Mapbox Layer

To prevent `CONTEXT_LOST_WEBGL` exceptions caused by creating multiple WebGL canvases, the platform injects a single Three.js scene directly into Mapbox GL JS's existing WebGL rendering loop using a custom layer interface:

```
import mapboxgl from 'mapbox-gl';
import * as THREE from 'three';

export class MassingThreeLayer implements
mapboxgl.CustomLayerInterface {
  id = 'building-massing-three-layer';
  type = 'custom' as const;
  renderingMode = '3d' as const;

  private camera: THREE.Camera = new THREE.Camera();
  private scene: THREE.Scene = new THREE.Scene();
  private renderer: THREE.WebGLRenderer | null = null;
  private map: mapboxgl.Map | null = null;

  constructor(private massingMeshes: THREE.Mesh[]) {}

  onAdd(map: mapboxgl.Map, gl: WebGLRenderingContext) {
    this.map = map;
    this.renderer = new THREE.WebGLRenderer({
      canvas: map.getCanvas(),
      context: gl,
      antialias: true
    });
    this.renderer.autoClear = false;

    const ambientLight = new THREE.AmbientLight(0xffffff, 0.8);
    this.scene.add(ambientLight);

    const dirLight = new THREE.DirectionalLight(0xffffff, 0.6);
    dirLight.position.set(50, 100, 50);
    this.scene.add(dirLight);

    this.massingMeshes.forEach(mesh => this.scene.add(mesh));
  }

  render(gl: WebGLRenderingContext, matrix: number[]) {
    const m = new THREE.Matrix4().fromArray(matrix);
    this.camera.projectionMatrix = m;
  }
}
```

```

this.renderer?.resetState();
this.renderer?.render(this.scene, this.camera);
this.map?.triggerRepaint();
}

onRemove() {
  this.scene.traverse((object) => {
    if (object instanceof THREE.Mesh) {
      object.geometry.dispose();
      if (Array.isArray(object.material)) {
        object.material.forEach((mat) => mat.dispose());
      } else {
        object.material.dispose();
      }
    }
  });
  this.renderer?.dispose();
  this.renderer = null;
}
}

```

Deliverable 8: Financial Feasibility Engine

The financial engine couples procedural layouts with development pro-forma models reflecting Indian institutional underwriting standards.

Mathematical Pro-Forma Equations

- **Gross Development Value (GDV):** $\text{GDV} = \sum_{i \in \text{Typologies}} \left(A_{\text{saleable}, i} \times P_{\text{base}, i} + N_{\text{parking}} \times P_{\text{stall}} + R_{\text{commercial_lease}} \right)$
- **Total Development Cost (TDC):** $\text{TDC} = C_{\text{land}} + C_{\text{statutory}} + C_{\text{hard_construction}} + C_{\text{soft_costs}} + C_{\text{finance}} + C_{\text{contingency}}$ Where:
 - $C_{\text{land}} = \text{Agreed Purchase Price} + \text{Stamp Duty (5\%--7\%)} + \text{Registration Fees}$.
 - $C_{\text{statutory}} = \text{Scrutiny Fees} + \text{Development Charges} + \text{Premium FSI Cost} + \text{TDR Purchases}$.
 - $C_{\text{hard_construction}} = \sum_k (\text{Built-Up Area}_k \times \text{Rate}_{\text{civ_finish}, k})$.
 - $C_{\text{finance}} = \sum_{t=1}^T \text{Debt Outstanding}_t \times \left(\frac{r_{\text{annual}}}{12} \right)$.
- **Developer Profit and Margins:** $\text{Gross Development Profit} = \text{GDV} - \text{TDC}$
 $\text{Return on Cost (RoC)} = \frac{\text{Gross Profit}}{\text{TDC}}, \quad \text{Gross}$

- Profit Margin} = \frac{\text{Gross Profit}}{\text{GDV}}
- **Break-Even Metrics:** \text{Break-Even Sales Price per sq ft} = \frac{\text{TDC} - (\text{Parking Revenue} + \text{Ancillary Revenue})}{A_{\text{saleable_total}}} \\ \text{Break-Even Saleable Area} = \frac{\text{TDC} - (\text{Parking Revenue} + \text{Ancillary Revenue})}{P_{\text{base_weighted}}}

Deliverable 9: AI + Regulation RAG Architecture

The platform strictly separates **deterministic computation** from **probabilistic AI inferences**. Generative design systems fail when they rely on Large Language Models (LLMs) to perform spatial math or extract numeric setback values directly. LLMs are prone to hallucinations, cannot reliably compute polygon intersections, and cannot guarantee statutory compliance. Deterministic engines handle all spatial math, geometric transformations, and financial formulas. The AI layer is restricted to parsing municipal planning documents, explaining regulatory rules, and powering natural-language interfaces.

- **Ingestion Pipeline:** Scans statutory planning documents (gazettes, bye-laws, circulars) using OCR models that preserve layout, section hierarchies, and tables.
- **Knowledge Representation:** Extracted clauses are structured into a property graph database (Neo4j). Nodes represent planning entities (Setback, RoadWidth, FSI_Incentive), and edges represent statutory dependencies (DEPENDS_ON, EXEMPTS, SUPERSEDES).
- **Human Verification Gate:** All extracted rules must be reviewed and verified by a licensed municipal architect before being published to the production rule engine.
- **Auditable Query Resolution:** When a user queries the platform ("Why are side setbacks set to 4.5\text{ m}?"), the copilot combines vector search with graph traversal to return the plain-text answer alongside an auditable legal citation: [Source: Maharashtra UDCPR 2020, Chapter 6, Table [span_143](start_span)[span_143](end_span)[span_144](start_span)[span_144](end_span)[span_145](start_span)[span_145](end_span)6-G, Clause 6.3(b)].

Deliverable 10: Database Schema

The platform uses PostgreSQL 16 with PostGIS extensions to manage cadastral boundaries, statutory constraints, architectural layouts, and immutable audit logs.

```
CREATE EXTENSION IF NOT EXISTS "uuid-osspl";
CREATE EXTENSION IF NOT EXISTS "postgis";
```

```
CREATE TYPE land_tenure_type AS ENUM (
    'PRIVATE_FREEHOLD', 'PRIVATE_LEASEHOLD', 'COLLECTOR_CLASS_II',
    'COLLECTOR_LEASEHOLD', 'GOVERNMENT_LEASEHOLD',
    'MHADA_REDEVELOPMENT',
    'MUNICIPAL_ALLOTMENT'
);
```

```
CREATE TYPE constraint_severity AS ENUM (
    'HARD_ABSOLUTE_EXCLUSION', 'CONDITIONAL_NOC_REQUIRED',
    'HEIGHT_RESTRICTION_SURFACE', 'SETBACK_OFFSET_MODIFIER'
```

```

);

CREATE TABLE parcels (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL,
    project_id UUID NOT NULL,
    cadastral_survey_number VARCHAR(128) NOT NULL,
    tenure_type land_tenure_type NOT NULL DEFAULT 'PRIVATE_FREEHOLD',
    srid_projected INT NOT NULL,
    boundary_wgs84 GEOMETRY(MultiPolygon, 4326) NOT NULL,
    boundary_projected GEOMETRY(MultiPolygon) NOT NULL,
    gross_area_sqm NUMERIC(14, 4) GENERATED ALWAYS AS
(ST_Area(boundary_projected)) STORED,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    version INT NOT NULL DEFAULT 1,
    is_deleted BOOLEAN NOT NULL DEFAULT FALSE,
    CONSTRAINT chk_boundary_wgs84_valid CHECK
(ST_IsValid(boundary_wgs84)),
    CONSTRAINT chk_boundary_projected_valid CHECK
(ST_IsValid(boundary_projected))
);

```

```

CREATE INDEX idx_parcels_boundary_wgs84 ON parcels USING GIST
(boundary_wgs84);
CREATE INDEX idx_parcels_boundary_projected ON parcels USING GIST
(boundary_projected);
CREATE INDEX idx_parcels_org_proj ON parcels(organization_id,
project_id) WHERE NOT is_deleted;

```

```

CREATE TABLE parcel_roads (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    parcel_id UUID NOT NULL REFERENCES parcels(id) ON DELETE CASCADE,
    road_name VARCHAR(256),
    existing_width_meters NUMERIC(6, 2) NOT NULL CHECK
(existing_width_meters > 0),
    proposed_width_meters NUMERIC(6, 2) CHECK (proposed_width_meters
>= existing_width_meters),
    road_centerline GEOMETRY(LineString, 4326),
    road_boundary_edge GEOMETRY(LineString, 4326) NOT NULL,
    is_primary_access BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_parcel_roads_edge ON parcel_roads USING GIST
(road_boundary_edge);

```

```

CREATE TABLE parcel_constraints (

```

```

    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    parcel_id UUID NOT NULL REFERENCES parcels(id) ON DELETE CASCADE,
    constraint_name VARCHAR(128) NOT NULL,
    constraint_category VARCHAR(64) NOT NULL,
    severity constraint_severity NOT NULL,
    buffer_meters NUMERIC(6, 2) NOT NULL DEFAULT 0.0,
    vertical_clearance_limit NUMERIC(8, 2),
    footprint_wgs84 GEOMETRY(Geometry, 4326) NOT NULL,
    footprint_projected GEOMETRY(Geometry) NOT NULL,
    buffer_geometry_projected GEOMETRY(Geometry) NOT NULL,
    statutory_reference VARCHAR(256),
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_parcel_constraints_geom ON parcel_constraints USING
GIST (buffer_geometry_projected);

CREATE TABLE parcel_trees (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    parcel_id UUID NOT NULL REFERENCES parcels(id) ON DELETE CASCADE,
    tag_number VARCHAR(64),
    species VARCHAR(128),
    trunk_girth_meters NUMERIC(4, 2),
    canopy_radius_meters NUMERIC(4, 2) NOT NULL DEFAULT 3.0,
    mandatory_root_protection_buffer NUMERIC(4, 2) NOT NULL DEFAULT
2.5,
    location_wgs84 GEOMETRY(Point, 4326) NOT NULL,
    location_projected GEOMETRY(Point) NOT NULL,
    is_felling_prohibited BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_parcel_trees_location ON parcel_trees USING GIST
(location_wgs84);

```

Deliverable 11: API Specification

The platform exposes RESTful endpoints supporting JSON serialization, Bearer token authentication, rate limiting, and idempotency headers.

Core Endpoint Specifications

- POST /api/v1/projects: Initializes a project record.
 - *Payload*: {"title": "Andheri Heights", "tenureType": "PRIVATE_FREEHOLD", "sridProjected": 32643}
 - *Response* (201 Created): {"projectId": "b2f6c52a-...", "status": "ACTIVE"}

- GET /api/v1/projects/:id: Retrieves complete parcel details, boundaries, active constraints, and feasibility records.
- PUT /api/v1/projects/:id/geometry: Debounced auto-save endpoint for updating parcel boundary coordinates, internal deductions, and tree markers.
 - *Payload*: GeoJSON FeatureCollection with updated geometries.
 - *Response* (200 OK): {"grossAreaSqm": 929.03, "isTopologyValid": true}
- POST /api/v1/projects/:id/feasibility/generate: Queues asynchronous generation of residential floor plates, structural grids, and parking layouts.
 - *Payload*: {"optimizationProfile": "MAX_CARPET_AREA", "parkingMode": "PODIUM"}
 - *Response* (202 Accepted): {"jobId": "job_9988", "status": "QUEUED"}
- GET /api/v1/projects/:id/reports/feasibility.pdf: Streams the compiled development feasibility prospectus as a binary PDF.

Deliverable 12: Frontend Architecture

The frontend is built as a single-page application using React 18, TypeScript, Tailwind CSS, TanStack Query, Zustand, and Mapbox GL JS.

- **Canvas Isolation**: The Mapbox GL JS map instance and drawing plugins are preserved in a React useRef rather than component state. This decouples the GPU rendering loop from React's virtual DOM, preventing re-renders from destroying the WebGL canvas context.
- **State Separation**: Ephemeral UI state (active tool modes, modal visibility) is managed in a client-side Zustand store. Server-synchronized data (parcel geometries, project lists, constraint forms) is managed via TanStack Query, providing automatic caching, re-fetching, and background synchronization.
- **Form Validation**: Form overlays use React Hook Form paired with Zod schemas, enforcing client-side type safety and preventing invalid inputs before submission.
- **Debounced Mutations**: Boundary drawing dispatches debounced updates through an AbortController pipeline, batching rapid vertex movements and cancelling obsolete in-flight requests to eliminate race conditions.

Deliverable 13: Testing Strategy

Quality assurance uses a multi-tier testing pipeline covering geometry math, municipal regulations, and end-to-end user workflows.

- **Geometric Precision Unit Tests**:
 - Validates Shoelace planar area against spherical geodesic excess on known test plots.
 - Asserts that self-intersecting bowtie polygons fail topology validation and are flagged by ST_IsValidReason.
 - Verifies that polygon buffering matches exact straight skeleton offsets without irregular corner rounding.
- **Regulatory Integration Tests**:
 - Asserts that an 18.0\text{ m} road width in Mumbai under DCPR 2034 yields a base FSI of 1.00, premium FSI of 0.50, and TDR of 0.90.
 - Asserts that an access road narrower than 6.0\text{ m} blocks high-rise residential

- layouts and flags an access violation.
- **Automated Canvas Testing (Playwright):**
 - Simulates browser user interactions: logs into the platform, geocodes a Mumbai parcel, digitizes a boundary on the canvas, and asserts that debounced auto-save triggers within 1000ms.
 - Visual regression testing compares WebGL canvas rendering against baseline pixel snapshots to catch UI rendering bugs.
- **Financial Pro-Forma Benchmarks:**
 - Cross-references pro-forma outputs (NPV, IRR, RLV) against audited Excel reference models, enforcing zero variance.

Deliverable 14: Security Architecture

The platform enforces enterprise-grade security to protect sensitive land acquisition data and commercial financial pro-formas.

- **Authentication & Session Management:** Replaces browser localStorage with secure, HttpOnly, SameSite=Strict cookies paired with automated refresh token rotation. This ensures tokens cannot be extracted by third-party scripts via cross-site scripting (XSS).
- **Multi-Tenant Data Isolation:** All database queries enforce PostgreSQL Row-Level Security (RLS), restricting access based on tenant organization IDs.
- **Content Security Policy (CSP):** Enforces strict HTTP CSP headers permitting connections only to authorized tile providers and API gateways: default-src 'self'; script-src 'self'; worker-src 'self' blob:; child-src blob:; connect-src 'self' https://api.mapbox.com https://*.tiles.mapbox.com; img-src 'self' data: blob: https://api.mapbox.com;
- **Data Encryption:** All data in transit is encrypted using TLS 1.3. Stored database volumes and S3 export assets are encrypted at rest using AES-256 with customer-managed keys.

Deliverable 15: MVP Feature Definition

The Minimum Viable Product (MVP) focuses on delivering core site feasibility workflows, postponing district-scale master planning and automated document parsing to later releases.

Feature Prioritization Matrix

Feature Module	Priority	Engineering Complexity	Business Value	Dependencies
WGS84 GeoJSON Boundary Drawing	P0	Moderate	Critical	Mapbox GL JS, Mapbox Draw
UTM Metric Planar Projection	P0	Moderate	Critical	PyProj, PostGIS Cartography
Negative Inward Setbacks & Offsetting	P0	High	Critical	CGAL / Shapely Straight Skeleton
Mumbai DCPR 2034 & UDCPR Rule Encoders	P0	High	Critical	Rule Schema, Gazette Ingestion

Feature Module	Priority	Engineering Complexity	Business Value	Dependencies
3D Massing Extrusions (Three.js)	P0	Moderate	High	WebGL Custom Layer, Mapbox GL JS
Fundamental Financial Pro-Forma (IRR, NPV)	P0	Moderate	Critical	Cash Flow Modeling Pipeline
Automated PDF Dossier Generator	P0	Low	High	Headless Chromium, PDFKit
Procedural Unit Packing (1BHK–3BHK)	P1	Very High	High	MILP Solver, NBC 2016 Dimensions
Subterranean Parking Stall Layout	P1	High	High	Ramp Gradient Math, Turning Radii
Bengaluru RMP 2015/2025 Ruleset	P1	Moderate	High	Zonal Amendments Ingestion
Multi-Objective NSGA-II Optimizer	P2	Very High	Medium	Genetic Algorithm Workpool
Vector BIM/CAD Exports (IFC, DXF)	P2	Moderate	Medium	IfcOpenShell, ezdxf Pipelines
Vision-Based Cadastral Survey Parsing	P3	Extreme	High	Vision LLM, Vectorization Models

Deliverable 16: 30/60/90 Day Development Roadmap

- **Days 1–30: Core Geometry & Statutory Foundations:**
 - Scaffold the React 18, TypeScript, and Tailwind CSS client application.
 - Deploy PostgreSQL 16 with PostGIS extensions on AWS RDS.
 - Build the Mapbox GL JS spatial canvas using useRef lifecycle isolation.
 - Implement straight skeleton setbacks, road widening, and tree buffer clipping using Shapely and CGAL.
 - Encode Mumbai DCPR 2034 Table 12 into the machine-readable rule engine.
- **Days 31–60: Architectural Layout, Massing, and Pro-Forms:**
 - Build the procedural building layout engine, generating vertical circulation cores, corridors, and units.
 - Inject Three.js 3D massing layers into Mapbox GL JS via custom WebGL layer pipelines.
 - Ingest Maharashtra UDCPR 2020/2025 and Bengaluru RMP 2015/2025 rulesets.
 - Implement the automated basement parking stall generator and access ramp slope

- calculator.
 - Build the core financial pro-forma engine and automated PDF report exporter.
- **Days 61–90: Optimization, AI Verification, and Beta Deployment:**
 - Deploy the multi-objective genetic layout solver (NSGA-II) for Pareto frontier exploration.
 - Build the RAG regulatory search service using Neo4j and vector embeddings with citation tracking.
 - Complete automated Playwright test suites and verify financial models against audited reference spreadsheets.
 - Conduct security hardening: implement HttpOnly cookie authentication and complete a SOC2 readiness review.
 - Launch an invite-only beta with five institutional real estate developers.

Deliverable 17: 6-Month and 12-Month Expansion Roadmap

- **Months 4–6: Jurisdictional Expansion & BIM Interoperability:**
 - Ingest statutory development regulations for Delhi NCR (MPD 2021), Hyderabad (G.O. Ms. 168), Chennai (TNCDDBR 2019), Kolkata (KMC Rules 2009/2025), and Jharkhand (JBBL 2016).
 - Implement native Industry Foundation Classes (IFC) and AutoCAD DXF vector export pipelines.
 - Add real-time collaborative map editing powered by WebSockets and CRDTs.
 - Build advanced Monte Carlo financial simulation modules for sensitivity and risk modeling.
- **Months 7–12: Vision Ingestion, Microclimates, and Regional Scale:**
 - Deploy an automated cadastral ingestion service that converts scanned revenue survey blueprints into vector GeoJSON.
 - Integrate WebGPU shaders for real-time solar radiation, daylight factor, and shadow simulations.
 - Expand the regulatory engine to support international jurisdictions (e.g., Dubai, Singapore, and the United Kingdom).

Deliverable 18: Team & Skill Requirements

Building the platform requires a multidisciplinary engineering and domain-expert team:

- **Chief Technology Officer / Principal Architect:** Oversees distributed systems, cloud infrastructure, and technical design.
- **Senior Computational Geometry Engineer:** Implements polygon clipping, straight skeleton offsets, Minkowski sums, and spatial decomposition using CGAL, Shapely, and C++.
- **Geospatial / PostGIS Data Engineer:** Manages cadastral boundaries, coordinate reference system projections, and spatial database optimizations.
- **Lead Municipal Architect & Urban Planner:** Translates statutory bye-laws, circulars, and zoning codes into machine-readable rule definitions.
- **Real Estate Financial Analyst:** Models institutional pro-formas, dynamic debt structures,

and equity distribution waterfalls.

- **Senior 3D WebGL / Frontend Engineer:** Builds interactive 3D massing visualizations and user interfaces using React, Mapbox GL JS, and Three.js.
- **Full-Stack Platform Engineers:** Implements backend microservices, event streams, API contracts, and user management systems.
- **QA / Automated Test Engineer:** Authors automated Playwright integration tests and verifies geometric precision across spatial features.

Deliverable 19: Comprehensive Risk Register

Risk ID	Risk Description	Severity	Likelihood	Mitigation Strategy
RSK-GEO-01	Degenerate Polygon Ingestion: User drawings with self-intersections or duplicate vertices crash clipping operations.	High	High	Run client-side geometric validation checks: auto-repair slivers via ST_MakeValid and reject invalid geometries.
RSK-REG-02	Statutory Rule Hallucination: AI layer invents or misquotes setback or FSI numbers, creating legal liability.	Critical	Low	Maintain strict separation: LLMs explain regulations, but all numbers must come from deterministic rule engines.
RSK-REG-03	Municipal Bye-Law Lag: Outdated rulesets fail to reflect recent government resolutions or gazette amendments.	High	Medium	Implement an administrative rule curation portal with automated gazette scrapers and mandatory sign-offs by local architects.
RSK-WGL-04	Browser WebGL Memory Leaks: Switching views exhausts browser WebGL contexts, crashing the map canvas.	High	Medium	Manage Mapbox and Three.js lifecycles using useRef, ensure complete cleanup on unmount, and share a single WebGL context.
RSK-SEC-05	Land Acquisition Data Leaks: Cross-tenant data	Critical	Low	Enforce PostgreSQL Row-Level

Risk ID	Risk Description	Severity	Likelihood	Mitigation Strategy
	breaches expose confidential land bids to competitors.			Security by organization ID, encrypt spatial data with tenant keys, and require multi-factor auth.
RSK-FIN-06	Construction Cost Inflation Errors: Flat construction cost estimates fail to capture market volatility, skewing pro-formas.	Medium	High	Model construction costs dynamically, linking project schedules to regional material indices and running Monte Carlo simulations.

Deliverable 20: Complete Worked Example

To validate the integration across the platform's geometric, regulatory, and financial engines, this walkthrough tracks an end-to-end feasibility analysis for a representative real-world parcel.

1. Parcel Input & Site Context

- **Location:** Andheri East, Mumbai Suburban District, Maharashtra, India.
- **Governing Regulation:** Development Control and Promotion Regulations for Greater Mumbai (DCPR 2034, Regulation 30, Table 12).
- **WGS84 Coordinates:** v_0 : [72.850000, 19.115000], v_1 : [72.850380, 19.115000], v_2 : [72.850380, 19.115226], v_3 : [72.850000, 19.115226].
- **Projected System:** UTM Zone 43N (WGS84, SRID 32643).
- **Dimensions:** Rectangular parcel measuring 40.00 m (frontage) $\times$ 23.225 m (depth).
- **Gross Plot Area (GPA):** 929.03 m^2 ($10,000.00\text{ sq ft}$).
- **Abutting Access Road Width:** 18.00 m (Southern boundary edge).
- **Statutory Road Widening Line:** 2.00 m reservation along the entire 40.00 m road frontage.
- **Preserved Tree:** One mature tree located at local coordinates $(30.00\text{ m}, 1.00\text{ m})$ with a mandatory 2.50 m root protection radius.

2. Spatial Deductions & Net Developable Envelope

- **Road Widening Area:** $A_{\text{widening}} = 40.00\text{ m} \times 2.00\text{ m} = 80.00\text{ m}^2 \quad (861.11\text{ sq ft})$
- **Net Plot Area (NPA):** $\text{NPA} = \text{GPA} - A_{\text{widening}} = 929.03\text{ m}^2 - 80.00\text{ m}^2 = 849.03\text{ m}^2 \quad (9,138.89\text{ sq ft})$
- **Mandatory Municipal Setbacks** (DCPR 2034 Table 12 & NBC 2016):
 - *Front Setback* (measured from post-widening road line): 4.50 m .
 - *Rear Setback*: 3.00 m .

- *Side Setbacks* (East and West): 3.00\text{ m} each.
- **Buildable Footprint:**
 - \text{Buildable Width} = 40.00\text{ m} - (3.00\text{ m} + 3.00\text{ m}) = 34.00\text{ m}.
 - \text{Buildable Depth} = (23.225\text{ m} - 2.00\text{ m}) - (4.50\text{ m} + 3.00\text{ m}) = 13.725\text{ m}.
 - \text{Gross Footprint Area} = 34.00\text{ m} \times 13.725\text{ m} = 466.65\text{ m}^2 \quad (5,022.98\text{ sq ft}).
- **Tree Root Buffer Deduction:** The circular root buffer (\text{Radius} = 2.50\text{ m}) overlaps the buildable footprint by 14.20\text{ m}^2. \text{Net Buildable Footprint Area} = 466.65\text{ m}^2 - 14.20\text{ m}^2 = 452.45\text{ m}^2 \quad (4,870.13\text{ sq ft})

3. FSI Allocation & Vertical Massing

Under DCPR 2034 Table 12 for an 18.00\text{ m} road in suburban Mumbai:

- \text{Base FSI} = 1.00
- \text{Premium FSI} = 0.50 (purchased at 50% of the Annual Statement of Rates [ASR])
- \text{TDR Loading} = 0.90 (purchased on open market)
- \text{Maximum Permissible FSI} = 1.00 + 0.50 + 0.90 = 2.40
- **Permissible Built-Up Area (BUA):** \text{Permissible BUA} = \text{NPA} \times \text{FSI} = 849.03\text{ m}^2 \times 2.40 = 2,037.67\text{ m}^2 \quad (21,933.34\text{ sq ft})
- **Vertical Stacking:**
 - *Ground Level:* Stilt parking, entrance lobby, and fire command center (4.20\text{ m} floor-to-floor height).
 - *Habitable Floors:* Target floor plate = 340.00\text{ m}^2 (3,659.73\text{ sq ft}), leaving space on the slab for balconies and services.
 - \text{Floor Count} = \frac{2,037.67\text{ m}^2}{340.00\text{ m}^2} = 6.0 \rightarrow 6\text{ Habitable Floors}.
 - \text{Total Height} = 4.20\text{ m} + (6 \times 3.05\text{ m}) + 1.20\text{ m} = 23.70\text{ m}.
 - Height is within the 32.0\text{ m} limit for this road class and satisfies NBC high-rise fire safety requirements (>15\text{ m}).

4. Floor Plate Planning and Unit Mix

Each of the 6 habitable floor plates (340.00\text{ m}^2) is laid out to optimize internal circulation and daylight access:

- *Vertical Core:* Central core with two 8-passenger elevators, two fire-rated staircases (1.50\text{ m} flight width), and an MEP utility duct. Total core area = 48.00\text{ m}^2.
- *Corridor & Circulation:* Central corridor measuring 1.80\text{ m} \times 12.00\text{ m} = 21.60\text{ m}^2.
- *Net Habitable Area per Floor:* 340.00\text{ m}^2 - (48.00\text{ m}^2 + 21.60\text{ m}^2) = 270.40\text{ m}^2 (2,910.56\text{ sq ft}).
- *Unit Mix per Floor:*
 - Two 2BHK units measuring 75.00\text{ m}^2 RERA carpet area each.
 - Two 1BHK units measuring 45.00\text{ m}^2 RERA carpet area each.

- Common interior partition walls and shafts: 30.40 m^2 .
- **Total Project Program:** 6 floors $\times$ 4 units/floor = **24 Apartments** (12 $\times$ 2 BHK, 12 $\times$ 1 BHK).
- **Total RERA Carpet Area:** $6 \times (2 \times 75.0 + 2 \times 45.0) = 1,440.00 \text{ m}^2$ (15,500.03 sq ft).
- **Saleable Area** (using a 38% super-built-up loading factor): $15,500.03 \times 1.38 = 21,390.04 \text{ sq ft}$ ($1,987.20 \text{ m}^2$).

5. Parking Stall Layout

- **Statutory Requirements** (DCPR 2034):
 - 2 BHK: 1 stall per unit $\rightarrow$ 12 stalls.
 - 1 BHK: 0.5 stalls per unit $\rightarrow$ 6 stalls.
 - Visitor Parking: 10% statutory surcharge $\rightarrow$ 2 stalls.
 - **Total Required:** 20 car stalls.
- **Parking Allocation:**
 - **Stilt Level:** 8 surface stalls + 10 two-wheeler bays.
 - **Basement Level:** 600.00 m^2 single-level basement accommodating 14 stalls, a 6.0 m two-way driveway, and an 1:8 access ramp.
 - **Total Provided:** **22 Car Stalls** (fully compliant).

6. Financial Pro-Forma Summary

A. CAPITAL REVENUES (GDV)

Residential Saleable Area: 21,390 sq ft @ INR 22,000 / sq ft = INR 47,05,80,000

Covered Car Parking Sales: 20 Allocated Stalls @ INR 5,00,000 = INR 1,00,00,000

 GROSS DEVELOPMENT VALUE (GDV) = INR 48,05,80,000

B. DEVELOPMENT COSTS (TDC)

1. Land Acquisition:

Agreed Land Price (9,138.89 sq ft Net Plot Area) = INR 18,00,00,000

Stamp Duty & Legal Charges (~7%)

[span_47] (start_span) [span_47] (end_span) = INR 1,26,00,000

2. Statutory Fees & FSI Premiums:

Premium FSI Purchase (0.50 FSI = 4,569 sq ft @ 50% ASR) = INR 2,28,45,000

TDR Purchase from Market (0.90 FSI = 8,225 sq ft) = INR 2,87,87,500

Municipal Development & Scrutiny Charges = INR 85,00,000

3. Hard Construction Costs:

Basement Construction (6,458 sq ft BUA @ INR 3,200/sq ft) = INR 2,06,65,600

Superstructure (25,000 sq ft GFA @ INR 2,800/sq ft) = INR 7,00,00,000

4. Soft, Finance & Contingency:

Architectural, Structural, Geo-Consultant Fees = INR 65,00,000

Marketing, Brokerage, and Sales Commissions (3% of GDV) = INR 1,44,17,400

Finance & Debt Servicing Charges = INR 2,40,00,000

Project Contingency Reserve (4% of Hard Costs) = INR 36,26,624

TOTAL DEVELOPMENT COST (TDC) = INR 39,19,42,124

C. PROFITABILITY & RETURNS

Gross Development Profit (GDV - TDC) = INR 8,86,37,876

Developer Return on Cost (RoC = Profit / TDC) = 22.61%

Gross Profit Margin (Profit / GDV) = 18.44%

Project Internal Rate of Return (IRR, 36-Month Horizon) = 24.85%

Net Present Value (NPV @ 12% Cost of Capital) = INR 3,94,20,540

7. Sensitivity Analysis Matrix

Construction Cost Variance \ Sales Realization Variance	Downside Case (-10% Realization: INR 19,800/sqft)	Base Case (INR 22,000/sqft)	Upside Case (+10% Realization: INR 24,200/sqft)
High Construction Cost (+10%)	Profit: INR 3,25,12,714 RoC: 8.11% IRR: 10.4%	Profit: INR 7,95,70,714 RoC: 20.03% IRR: 21.8%	Profit: INR 12,66,28,714 RoC: 31.87% IRR: 32.1%
Base Construction Cost (INR 2,800/sqft)	Profit: INR 4,15,79,876 RoC: 10.61% IRR: 13.2%	Profit: INR 8,86,37,876 RoC: 22.61% IRR: 24.8%	Profit: INR 13,56,95,876 RoC: 34.62% IRR: 35.4%
Low Construction Cost (-10%)	Profit: INR 5,06,47,038 RoC: 13.23% IRR: 16.1%	Profit: INR 9,77,05,038 RoC: 25.52% IRR: 27.9%	Profit: INR 14,47,63,038 RoC: 37.81% IRR: 38.6%

Deliverable 21: Final "Build From Zero" Technical

Blueprint

This section synthesizes the system architecture into a production deployment blueprint for engineering teams building the platform from scratch.

Component Integration and Communication Topology

- **Presentation Layer:** The React 18 frontend renders the interactive Mapbox GL JS map canvas, heads-up metrics display, and dynamic constraint forms. The Mapbox instance and its drawing controllers are managed within React useRef hooks to prevent component re-renders from re-initializing the WebGL canvas.
- **Transport & Security Gateway:** The client connects to the backend over HTTPS and WebSockets via a reverse proxy and API Gateway (Kong). Outgoing requests authenticate using secure, HttpOnly, SameSite=Strict cookies with automated token refresh cycles, protecting user sessions from XSS token theft.
- **Project & User Management Service:** A Go microservice handles tenant authentication, role-based access control, project metadata, and cadastral revision history.
- **Spatial Compute Service:** A Python microservice built on Shapely, PyProj, and CGAL processes spatial geometries. It normalizes WGS84 GeoJSON into local metric planar projections (UTM), repairs invalid geometries, and computes straight skeleton setbacks, road widening deductions, and tree buffer exclusions.
- **Event-Driven Asynchronous Processing:** Long-running computational tasks—such as generative floor plate layouts, parking bay generation, and multi-objective optimization—are published to an Apache Kafka message bus. A pool of background workers consumes these jobs, computes solutions, and updates project state asynchronously.
- **Regulatory & Financial Microservices:**
 - The **Regulation Engine** evaluates parcel attributes against versioned municipal bye-law schemas (e.g., DCPR 2034, UDCPR), returning statutory FSI caps and setback requirements.
 - The **Layout Engine** synthesizes structural grids, circulation corridors, and residential units within the buildable envelope.
 - The **3D Massing Engine** extrudes floor plates into parametric building tiers rendered via Three.js custom layers inside Mapbox.
 - The **Financial Engine** pairs massing configurations with pro-forma cash flows, computing IRR, NPV, and residual land values.
- **Persistence & Object Storage Tier:** All spatial boundaries, municipal rulesets, and revision histories are stored in PostgreSQL 16 with PostGIS spatial extensions. Real-time metrics and active job states are cached in Redis. Exported CAD files, BIM models, and generated PDF feasibility prospectuses are stored in AWS S3 with signed access URLs.
- **Client Synchronization:** When background generation completes, workers push the results to the client over WebSockets. The React application caches the data in TanStack Query and renders updated geometries, 3D massings, and pro-forma metrics on the canvas.

This decoupled architecture allows computational components—such as the regulatory rule engine or the 3D massing generator—to be extended, tested, and scaled independently. By grounding generative layouts in deterministic spatial geometry and versioned statutory

bye-laws, the platform provides real estate developers, master planners, and institutional investors with an enterprise-grade system for rapid site feasibility and land development intelligence.

Works cited

1. Government of Jharkhand - Ranchi Municipal Corporation, https://www.ranchimunicipal.com/Downloads/10102017191852546_DownloadFiles.pdf
2. Cluster Redevelopment in Mumbai (DCPR 33(9)), <https://societyredevelopmentmumbai.com/government-policies/cluster-redevelopment/>
3. Zonal Regulation Amendment 2015 Dt05/01/2026 Final - Scribd, <https://www.scribd.com/document/988571121/Zonal-Regulation-Amendment-2015-Dt05-01-2026-Final>
4. Updated-UDCPR-30.01.25.pdf - PMRDA Pune, <https://www.pmrda.gov.in/wp-content/uploads/2025/09/Updated-UDCPR-30.01.25.pdf>
5. Delhi Master Plan 2021 Overview | PDF - Scribd, <https://www.scribd.com/document/72070511/Master-Plan-2021-Copy>
6. UDCPR Maharashtra 2025 Regulations | PDF | Parking - Scribd, <https://www.scribd.com/document/868547832/UDCPR-Updated-30-01-25>
7. BFS_Rule_2021_English.pdf - Fire NOC & Audit Portal, https://firenoc.bihar.gov.in/noc/act/BFS_Rule_2021_English.pdf
8. Bye laws fire | PDF - Slideshare, <https://www.slideshare.net/slideshow/bye-laws-fire/146023787>
9. National Building Code 2016 Overview | PDF | Fire Sprinkler System, <https://www.scribd.com/document/562683387/NBC-codes>
10. AI Floor Plan Generator — Plot-to-Plan + Sketch-to-Plan AI for India, <https://www.studiomatrix.org/guides/ai-floor-plan-generator>
11. Premium FAR Regulations for Bengaluru | PDF | Government - Scribd, <https://www.scribd.com/document/846079197/Premium-FAR-Gazette>
12. Jharkhand Building Bye Laws 2016 | PDF - Scribd, <https://www.scribd.com/document/436563666/Jharkhand-Building-Bye-Laws-2016>
13. Fire NOC 2026 | Provisional & Final Under NBCS 2026, State Acts, <https://www.factorycompliance.in/fire-noc>
14. Introduction to National Building Code and Steps in a Construction, <https://skill-lync.com/student-projects/introduction-to-national-building-code-and-steps-in-a-construction-project-136>
15. TNCDBR_Rules,_2019.pdf - ccmc, https://ccmc.gov.in/img/upload/TNCDBR_Rules,_2019.pdf
16. Jharkhand Building Bye-laws 2021 Update | PDF | Gravel - Scribd, <https://www.scribd.com/document/911868681/Amendment-Viii-of-Jbbl-2016>
17. GBA NEW NOTIFICATION DATED 04-06-2026* 1. GBA ALLOWS, <https://www.facebook.com/planapprovalonline/posts/gba-new-notification-dated-04-06-20261-gba-allows-only-10-voids-the-total-permis/1109227501914511/>
18. TestFit Reviews in 2026 - SourceForge, <https://sourceforge.net/software/product/TestFit/>
19. THE JHARKHAND GAZETTE, <https://avantiscdnprodstorage.blob.core.windows.net/legalupdatedocs/55742/Jharkhand-Building-XIth-Amendment-Bye-laws-2026-MAY112026.pdf>
20. Jharkhand Building (XIth Amendment) Bye-laws, 2026, <https://teamleaseregtech.com/updates/article/55742/jharkhand-building-xith-amendment-bye-laws-2026/>
21. FSI in Bangalore Real Estate: Meaning, Full Form & Rules 2026, <https://alldoors.in/blogs/fsi-in-bangalore-real-estate>
22. AI Tools Directory for Architects & AEC |

Archomation, <https://www.archomation.com/ai-tools> 23. Updated KMC Building Rules 2025 | PDF | Indemnity - Scribd, <https://www.scribd.com/document/1004694752/Updated-KMC-Building-Rules-2025> 24. Building Setbacks Across India — A State-by-State Guide to, <https://www.studiomatrix.org/guides/setbacks-across-india>